# David Mayowa Onaiyekan

Machine Learning Engineer

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## Profile

Machine Learning Engineer with an M.Sc. in Artificial Intelligence and hands-on research experience in Deep Learning, Computer Vision, and Generative AI. Experienced in training and evaluating Vision Transformers and CNN-based architectures to improve model performance on real-world datasets. Proficient in PyTorch, representation learning, and deploying AI solutions with Docker and AWS. Extends this foundation with production-grade AI engineering experience building RAG systems, agentic AI workflows, and full-stack automation pipelines.

## Skills

- **ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **Computer Vision:** OpenCV, Image Processing, CNNs, Vision Transformers, OCR, GANs
- **LLMs & AI Engineering:** OpenAI, Anthropic, Llama, Hugging Face, LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, MCP
- **Databases & Vector Databases:** PostgreSQL, MySQL, MongoDB, Qdrant, Pinecone, ChromaDB
- **MLOps, Cloud & Deployment:** AWS, Docker, FastAPI, CI/CD, GitHub Actions
- **Automation & Workflows:** n8n, Make, AI Workflow Orchestration
- **Languages:** Python (Advanced), JavaScript, TypeScript

## Education

**Master of Science in Artificial Intelligence** **01/2026**
*Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany*

**Bachelor of Technology in Computer Science** **11/2018**
*Federal University of Technology, Minna, Nigeria*

## Research Experience

**Machine Learning Researcher** **06/2025 – 01/2026**
*Pattern Recognition Lab, FAU* *Erlangen, Germany*

- Fine-tuned and evaluated alternative backbone architectures for the AFFGAN handwriting style encoder to improve capture of writer-specific handwriting features, benchmarking against the original baseline.
- Extended the system beyond word-level synthesis to line-level handwriting generation, integrating and adapting an existing line-generation architecture to support full-sentence output.
- Deployed the handwriting generation pipeline as a web application, enabling users to select a writing style, input custom text, and instantly generate a corresponding handwriting image.
- Integrated TrOCR into a teacher–student distillation framework to evaluate and improve the readability of generated handwriting, improving OCR accuracy by 20% over the baseline.

**Tech:** PyTorch, GANs, CNN, ViT, TrOCR, Streamlit, CUDA

**Computer Vision Researcher** **12/2023 – 01/2025**
*Institute for Factory Automation and Production Systems (FAPS)* *Erlangen, Germany*

- Collaborated within a research team to execute systematic data cleaning on thousands of industrial coil-winding images, leveraging Cleanlab Studio's autofix to establish a high-fidelity baseline for downstream model training.
- Trained and evaluated fully-supervised baseline classifiers (DINOv2-L and EfficientNetV2s) to benchmark against semi-supervised (FixMatch/MixMatch) gains across four label-fraction splits (10%,

25%, 50%, 100%).
- Ran systematic hyperparameter optimization with Optuna across backbone/algorithm combinations, tracking all trials via a persistent SQLite results database for reproducibility.
- Designed and implemented a config-driven multi-label SSL training pipeline supporting FixMatch and MixMatch, achieving a 90% macro-F1 average score with FixMatch.

**Tech:** PyTorch, DINOv2-L, EfficientNetV2-L, FixMatch, Optuna, WandB, Docker

## AI Engineering Projects

**Smart Dispatcher — Agentic Ticket Triage & Dispatch** **04/2026 – 06/2026**

- Developed and deployed a production-ready agentic AI system that automatically triages support tickets and generates primary and fallback dispatch plans, reducing manual decision-making while keeping humans in the approval loop.
- Designed a 6-node LangGraph workflow with a self-correction loop that detects failed repair plans and automatically retrieves firmware-based workarounds via RAG, without additional LLM prompting.
- Engineered guardrails into the RAG retrieval pipeline — metadata-filtered vector search and regex-based extraction guards — to eliminate embedding confusion between similar error codes and false-positive part requirements, validated with 17 evaluation cases.
- Built a React/FastAPI dashboard with real-time SSE streaming that surfaces the agent's decision-making process node-by-node.
- Integrated the full agent pipeline with LangSmith observability — tracing every LLM call, tool invocation, and self-correction retry with per-step latency, token usage, and cost — to monitor and evaluate agent performance over time.

**Tech:** Python, LangGraph, Claude (Anthropic SDK), ChromaDB, FastAPI, LangSmith, Google Sheets API, Railway, Tailwind, Pydantic

**RAG-Powered Document Chat System** **02/2026 – 05/2026**

- Developed a production-deployed RAG application enabling users to upload PDFs and receive streaming, grounded answers with page-level source citations.
- Implemented a hybrid retrieval pipeline combining dense embeddings and BM25 keyword search, fused via Reciprocal Rank Fusion and reranked by Claude to improve answer grounding.
- Implemented RAGAS-based evaluation measuring faithfulness, answer relevancy, and correctness, streamed live into the UI.

**Tech:** Python, FastAPI, LlamaIndex, RAG, Qdrant, Docker

**Full-Stack Budget & Expense Tracker** **01/2026 – 03/2026**

- Developed and deployed a full-stack personal finance tracker that allows users to record transactions, monitor weekly spending limits, and calculate monthly savings dynamically.
- Implemented secure user authentication and session management with Supabase Auth, scoping all budget and transaction data to the authenticated user.
- Built a RESTful backend using Next.js API routes and Prisma ORM to manage transaction creation, deletion, and budget data stored in a PostgreSQL database.
- Built a responsive dashboard with real-time financial summaries and deployed using Vercel with a Supabase PostgreSQL database.
- Designed interactive data visualizations with Recharts, including weekly spend-vs-limit and category breakdown charts, to help users spot overspending at a glance.

**Tech:** TypeScript, Next.js, Prisma ORM, PostgreSQL, Supabase, Supabase Auth, Recharts, Vercel

## Languages

English (Native) — German (Intermediate)